

DNS Internal Configuration

1. Open Default Zones File

```
sudo nano /etc/bind/named.conf.default-zones
```

2. At the bottom of the file add this text

Change < t > to your team number

```
zone "team<t>.org" IN {
    type master;
    file "/etc/bind/zones/forward.team<t>.org";
    allow-update { none; };
};

zone "<t>.168.192.in-addr.arpa" IN {
    type master;
    file "/etc/bind/zones/reverse.team<t>.org";
    allow-update { none; };
};
```

To save and exit the file do the following **CTRL+X** -> then press **Y** -> then press **ENTER**

3. Create Directory/Folder to Contain Zones

```
sudo mkdir /etc/bind/zones
```

```
sudo cp /etc/bind/db.empty /etc/bind/zones/forward.team<t>.org
sudo cp /etc/bind/db.empty /etc/bind/zones/reverse.team<t>.org
```

4. Configure hostname

```
sudo hostnamectl set-hostname ncae-lab2026
```

5. Configure forward.team.org

```
sudo nano /etc/bind/zones/forward.team<t>.org
```

Add/modify the following text to the file

```
$TTL      86400
@         IN      SOA      ncae-lab2026.team<t>.org root.ncae-lab2026. (
                               2          ; Serial
                               604800    ; Refresh
                               86400     ; Retry
                               2419200   ; Expire
                               86400    ) ; Negative Cache TTL
;
@         IN      NS       ncae-lab2026
ncae-lab2026 IN    A       192.168.<t>.12
www       IN      A       192.168.<t>.5
```

6. Configure reverse.team.org

```
sudo nano /etc/bind/zones/reverse.team<t>.org
```

Add/modify the following text to the file

```
$TTL      86400
@         IN      SOA      ncae-lab2026.team<t>.org. root.team<t>.org. (
                                2          ; Serial
                                604800    ; Refresh
                                86400     ; Retry
                                2419200   ; Expire
                                86400 ) ; Negative Cache TTL
;
@         IN      NS       ncae-lab2026.
5         IN      PTR      www.team<t>.org
```

7. Restart Bind Server

```
sudo systemctl daemon-reload
sudo systemctl restart bind9
sudo systemctl status bind9
```

8. Configure PortForwarding for DNS (DST)

Navigate in Firefox to `http://192.168.<t>.1`

Login is:

- **username:** admin
 - **password:** NCAE
- After login , go to **Advanced** → **IP** → **Firewall** → **NAT** → **New**

Enter in the General Tab Section:

```
- Chain: dstnat
- Protocol: udp
- Dst. Port: 53
- In. Interface: WAN
```

Enter in the Action Tab Section:

```
- Action: dst-nat
- To Addresses: 192.168.t.12
- To Ports: 53
```

Select `OK`

8. Configure PortForwarding for DNS (Firewall)

Go to **Advanced** → **IP** → **Firewall** → **Filter Rules** → **New**

Enter in the General Tab Section:

```
- Protocol: udp
- Dst. Port: 53
- Out. Interface: LAN
```

Select OK

9. Configure PortForwarding for Web (DST)

Navigate in Firefox to `http://192.168.<t>.1`

Login is:

- **username:** admin
- **password:** NCAE

After login , go to **Advanced** → **IP** → **Firewall** → **NAT** → **New**

Enter in the General Tab Section:

```
- Chain: dstnat
- Protocol: tcp
- Dst. Port: 80
- In. Interface: WAN
```

Enter in the Action Tab Section:

```
- Action: dst-nat
- To Addresses: 192.168.<t>.5
- To Ports: 53
```

Select OK

10. Configure PortForwarding for DNS (Firewall)

Go to **Advanced** → **IP** → **Firewall** → **Filter Rules** → **New**

Enter in the General Tab Section:

```
- Protocol: udp
- Dst. Port: 80
- Out. Interface: LAN
```

Select OK

11. Test Internal DNS

On the web machine, navigate to <http://www.team<t>.org> in firefox

Or

```
curl http://www.team<t>.org
```

DNS ExternalConfiguration

12. Add External Zones

```
sudo cp /etc/bind/db.empty /etc/bind/zones/forward.team<t>.cnyhackathon.org
sudo cp /etc/bind/db.empty /etc/bind/zones/web.team<t>.cnyhackathon.org
```

13. Configure forward.team<t>.cnyhackathon.org

```
sudo /etc/bind/zones/forward.team<t>.cnyhackathon.org
```

Add/modify the text below

```
$TTL      86400
@         IN      SOA      ns1.team<t>.cnyhackathon.org root.ns1. (
                        2      ; Serial
                        604800 ; Refresh
                        86400  ; Retry
                        2419200 ; Expire
                        86400 ) ; Negative Cache TTL
;
@         IN      NS       ns1
ns1       IN      A        172.18.13.<t>
www       IN      A        172.18.13.<t>
```

14. Configure web.team.cnyhackathon.org

```
sudo /etc/bind/zones/web.team<t>.cnyhackathon.org
```

```
$TTL      86400
@         IN      SOA      ns1.team<t>.cnyhackathon.org. root.cnyhackathon.org. (
                        2      ; Serial
                        604800 ; Refresh
                        86400  ; Retry
                        2419200 ; Expire
                        86400 ) ; Negative Cache TTL
;
@         IN      NS       ns1.
2         IN      PTR      ns1.team<t>.cnyhackathon.org
2         IN      PTR      www.team<t>.cnyhackathon.org
```

15. Add new zones to the default-zones file

```
sudo /etc/bind/named.conf.default-zones
```

Scroll to the bottom of the file and add the following entries

```
zone "team<t>.cnyhackathon.org" IN {
    type master;
    file "/etc/bind/zones/forward.team<t>.cnyhackathon.org";
    allow-update { none; };
};

zone "13.18.172.in-addr.arpa" IN {
    type master;
    file "/etc/bind/zones/web.team<t>.cnyhackathon.org";
    allow-update { none; };
};
```

16. Restart Bind Server

```
sudo systemctl daemon-reload
sudo systemctl restart bind9
sudo systemctl status bind9
```

17. Browse to someone else's network

Add a secondary DNS server

a.

```
nmcli connection modify "Wired Connection 1" ipv4.dns "192.168.t.12 172.18.13.t"
```

b.

Or

The Top right-hand corner of the machine click the power button widget area. Then:

Click the > on Wired -> Wired Settings -> select the gear icon on the Wired Box -> IPV4

c.

Where 192.168.<t>.12 is add a comma and add someone else public IP

Example

```
192.168.2.12, 172.18.13.2
```

18. Browse to someone else's network

Final test perform the following queries

```
dig www.team<t>.org
dig -x 192.168.<t>.5
```

```
dig www.team<t>.cnyhackathon.org @172.18.13.<t>
dig -x 172.18.13.<t> @172.18.13.<t>
```